# Vadim Vadanici

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## Education

**Imperial College London** — London, UK

*MEng Biomedical Engineering (Computational Bioengineering Stream)* — *Expected 2028*

- Predicted First Class Honours; top of a 120+ cohort in Year 1 Mechanics & Electronics (79% overall, 83% Mechanics).
- Undergraduate Researcher, Dario Farina Lab (2025–2026): machine learning on EMG hand-gesture recordings.
- Year 3 modules: Statistical Signal Processing & Inference; Image Processing; Software Engineering for Bioengineers.

**École Active Bilingue Jeannine Manuel** — Paris, France

*French Baccalaureate, International Option (English)* — *2024*

- Highest distinction (17.61/20) — perfect scores in Mathematics, AP Calculus BC, History, English and AP Language.
- Honourable Mention, French Mathematics Olympiad; Concours Général candidate (Philosophy, Physics-Chemistry).

## Experience

**PITOR — Paris Institute for Transplantation & Organ Regeneration** — Paris, France

*Research Intern* — *August 2026–Present*

- Built a local LLM pipeline automating the thematic analysis of 6,000+ free-text responses from kidney-transplant patients, combining structured coding, embedding-based clustering and source-verified quotations.
- Benchmarked the clinical reasoning of frontier LLMs on Banff rejection diagnosis from lesion scores: ∼30 runs over all 240 biopsies across temperatures, reasoning modes and model versions, scored against expert pathologists.
- Preparing a first-author manuscript, continuing part-time alongside my degree.

**Okeiro (AI Medtech Start-up, Chronic Kidney Disease)** — Paris, France

*Summer Consulting Intern* — *June–July 2025*

- Designed a remote patient monitoring service model for kidney disease and pitched it to venture capital investors.

## Projects

**Neural Manifold Learning for Brain-Imaging Biomarkers — Imperial Group Project** — London, UK

*Project Manager (7-member team), supervised by Prof Simon Schultz* — *October 2026–Present*

- Year-long research project applying neural manifold learning to fMRI data from the UK PREVENT Dementia cohort.
- Refining MIMI, an information-theoretic manifold method, to find biomarkers of Alzheimer's-related circuit change.

**London AI x Science Hackathon (Iterate) — Polaron & Cognition Track** — London, UK

*One of 150 participants selected from 1,000+ applicants* — *October 2026*

- Segmented battery-electrode microstructure via multi-Otsu to predict manufacturing changes before batteries are built.
- Linked literature-based markers that differed significantly between batches to a Doyle–Fuller–Newman battery model.
- Interpreted a blind hold-out test, catching a leak: the imaging session was confounded with the batch.

**Job OS — Personal AI Job-Search Platform**

*TypeScript, Next.js, PostgreSQL, Claude API* — *September 2026–Present*

- Built a job-matching platform that ranks roles against my profile and fact-checks every AI-generated claim.

**Smart Elastic Resistance Bands for Physiotherapy — Imperial Group Project** — London, UK

*Project Manager (8-member team)* — *October 2025–June 2026*

- Designed sensor wristbands that calibrate to each patient's range of motion, count reps and gamify physiotherapy.
- Sole developer of the ReQuest software: the Bluetooth-driven game, plus a patient app and clinician dashboard (prescriptions, adherence tracking, messaging) on a FastAPI backend.
- Led the team to the Year 2 Biomedical Engineering Prize for Best Group Project.

## Certifications & Skills

**University of Cambridge — Machine Learning & AI for Medicine Summer School** (van der Schaar Lab) *2025*

**Optiver & Imperial College London — Trading Academy** *2026*

Month-long course on options theory and market making; implemented trading strategies in Python.

**Programming:** Python (NumPy, pandas, SciPy, scikit-learn, PyTorch), SQL, MATLAB, Java, Git, LaTeX

**Languages:** Fluent in English, French and Romanian; German (B1)

## Leadership & Extracurricular Activities

- Selected as one of 500 attendees from 5,000+ applicants for Y Combinator Startup School Paris (2026).
- Organiser, Nature Conference *Redefining Healthcare in the Age of AI* (Sorbonne, Paris, 2026).
- Imperial Neuroscience Society — selected for a mentorship programme led by neurotech founders.
- Former school basketball captain (10+ years); Imperial Basketball Team; EDUCAP/CAPSAA inclusive-sport organiser.
- Co-founder of a Quantum Mechanics club; member of the ENS Paris Ulm Maths Club.
- Three exhibitions with Duchamp Prize winner Tarik Kiswanson; subject of his work *Vadim* (kadist.org/work/vadim).
- **Interests:** Artificial intelligence, neuroscience, chess, poker, music (DJing, piano), skiing, football, philosophy.